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Course: Data Acquisition & Data Preparation

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Project: Predicting Survival Outcomes Among Lung Cancer Patients Using Machine Learning

1. Dataset Source and Description

The dataset was obtained from Kaggle: "Lung Cancer Prediction" by Rashad Mammadov. It contains 23,658 patient records and 38 variables covering demographic, clinical, treatment, comorbidity, laboratory, and survival information.

The prediction target is a binary survival outcome derived from `Survival_Months` using a 60-month cutoff:

- Short-Term Survivor: `Survival_Months < 60` → 11,710 patients (49.5%)
- Long-Term Survivor: `Survival_Months ≥ 60` → 11,948 patients (50.5%)

The dataset is well-balanced across classes and is suitable for binary classification.

2. Data Quality Assessment

Check	Result
Missing values	0 across all 38 variables
Duplicate rows	0
Duplicate Patient_IDs	0
Data type issues	None
Inconsistent categorical values	None detected

No imputation or deduplication was required.

3. Summary Statistics

`df.describe().T` was applied to all numerical variables. Key observations:

- All numerical features fall within clinically plausible ranges.
- No constant or near-constant columns.
- No variables exhibited extreme skew requiring transformation.

4. Missing Value Analysis

Every variable reported Missing_Count = 0 and Missing_Percentage = 0.0%. No missing-value treatment was performed — fabricating imputation when no data is missing would be methodologically incorrect.

5. Outlier Detection

The IQR method ($1.5 \times \text{IQR}$ rule) was applied to all numerical variables. Zero outliers were identified. No records were removed. Outlier detection was documented rather than "fixed" — the correct procedure when no outliers exist.

6. Data Cleaning Procedures

1. Removed Patient_ID — an identifier with no predictive value.
2. Removed Survival_Months from predictors — it was used to construct the target, so retaining it would cause target leakage.
3. Confirmed 0 missing values and 0 duplicates post-cleaning.

7. Feature Encoding Methods

Categorical predictor variables were one-hot encoded using `pd.get_dummies(drop_first=True, dtype=int)`:

- Encoding was fitted on the training set only, then applied to the test set.
- Test columns were reindexed to match training columns exactly.
- `drop_first=True` avoids multicollinearity (dummy variable trap).

8. Feature Scaling Methods

Numerical features were standardized using `StandardScaler` (mean = 0, std = 1):

- The scaler was fitted on the training set only.
- The same transformation was applied to the test set.
- This prevents information leakage from the test set into the training pipeline.

The train/test split was performed before encoding and scaling, using an 80/20 stratified split (`random_state=42`) to preserve class proportions.

9. Final Cleaned Dataset Description

- Rows: 23,658
- Columns: 30+ (after one-hot encoding)
- Missing values: 0
- Duplicates: 0

- Target: Survival_Group (Short-Term / Long-Term)
- File: data/lung_cancer_cleaned.csv

10. Reflection on Preprocessing Decisions

The most consequential decisions were (1) removing Survival_Months to prevent target leakage, and (2) performing the train/test split before encoding and scaling. Both are standard best practices but easy to overlook. There were absence of missing values and outliers, the correct response was to document that finding rather than force unnecessary preprocessing steps. One-hot encoding was considered over label encoding because all categorical variables are nominal (no inherent ordering). Standard Scaler was chosen because downstream models are distance-based (logistic regression, SVM). The resulting cleaned dataset is directly usable in the modeling stage without further transformation.